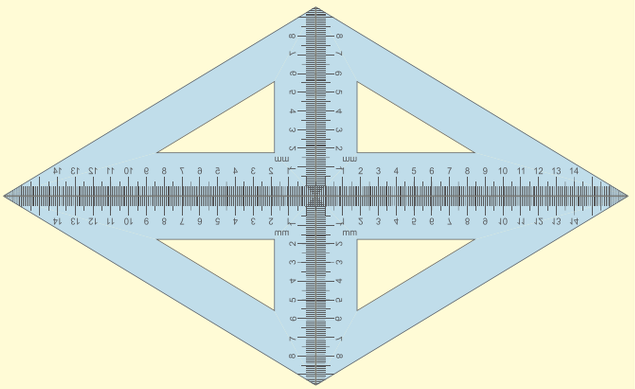
## Activity

4. We again use four identical 30°–60°–90° set- squares for this activity. Note carefully how they are placed touching one another.



   (i) Do we get a parallelogram now?

   (ii) What can we say about its lengths of sides, angles and diagonals?

   (iii) What is special about their diagonals?

Based on the outcome of the above activities, we can list out the various properties of the above quadrilaterals, all of which happen to be parallelograms!

| Special Quadrilaterals | All sides | All angles | Opposite Sides | | All angles | Opposite angles | Diagonals | |
|---|---|---|---|---|---|---|---|---|
| | Equal | Equal | Equal | Parallel | 90° | Supple-mentary | Bisect each other | Cut at rt.angles |
| Parallelogram | Sometimes | Sometimes | Always | Always | Sometimes | Sometimes | Always | Sometimes |
| Rhombus | Always | Sometimes | Always | Always | Sometimes | Sometimes | Always | Always |
| Rectangle | Sometimes | Always | Always | Always | Always | Always | Always | Sometimes |
| Square | Always | Always | Always | Always | Always | Always | Always | Always |



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